performance of the filter. This is also why floating point representation is typically used to implement windowed-sinc filters. Integers usually don't have enough dynamic range to capture the large variation of values contained in the filter kernel. How does the windowed-sinc filter perform in the time domain? Terrible! The step response has overshoot and ringing; this is *not* a filter for signals with information encoded in the time domain.

Examples of Windowed-Sinc Filters

An electroencephalogram, or EEG, is a measurement of the electrical activity of the brain. It can be detected as millivolt level signals appearing on electrodes attached to the surface of the head. Each nerve cell in the brain generates small electrical pulses. The EEG is the combined result of an enormous number of these electrical pulses being generated in a (hopefully) coordinated manner. Although the relationship between thought and this electrical coordination is very poorly understood, different frequencies in the EEG can be identified with specific mental states. If you close your eyes and relax, the predominant EEG pattern will be a slow oscillation between about 7 and 12 hertz. This waveform is called the *alpha rhythm*, and is associated with contentment and a decreased level of attention. Opening your eyes and looking around causes the EEG to change to the *beta rhythm*, occurring between about 17 and 20 hertz. Other frequencies and waveforms are seen in children, different depths of sleep, and various brain disorders such as epilepsy.

In this example, we will assume that the EEG signal has been amplified by analog electronics, and then digitized at a sampling rate of 100 samples per second. Acquiring data for 50 seconds produces a signal of 5,000 points. Our goal is to separate the alpha from the beta rhythms. To do this, we will design a digital low-pass filter with a cutoff frequency of 14 hertz, or 0.14

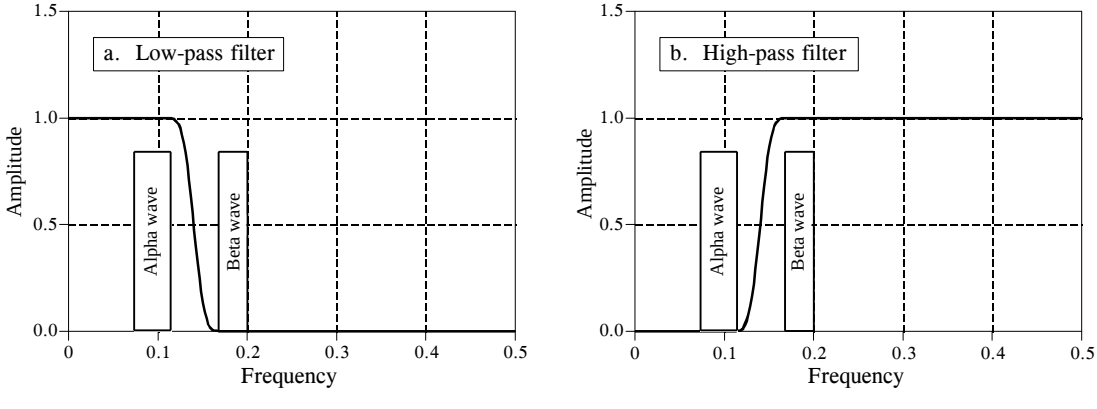

FIGURE 16-5
Example of windowed-sinc filters. The alpha and beta rhythms in an EEG are separated by low-pass and high-pass filters with $M = 100$. The program to implement the low-pass filter is shown in Table 16-1. The program for the high-pass filter is identical, except for a *spectral inversion* of the low-pass filter kernel.